CampusFlex

Use-Case Diagram and Fully Dressed Use Cases

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CampusFlex — Use Case Diagram

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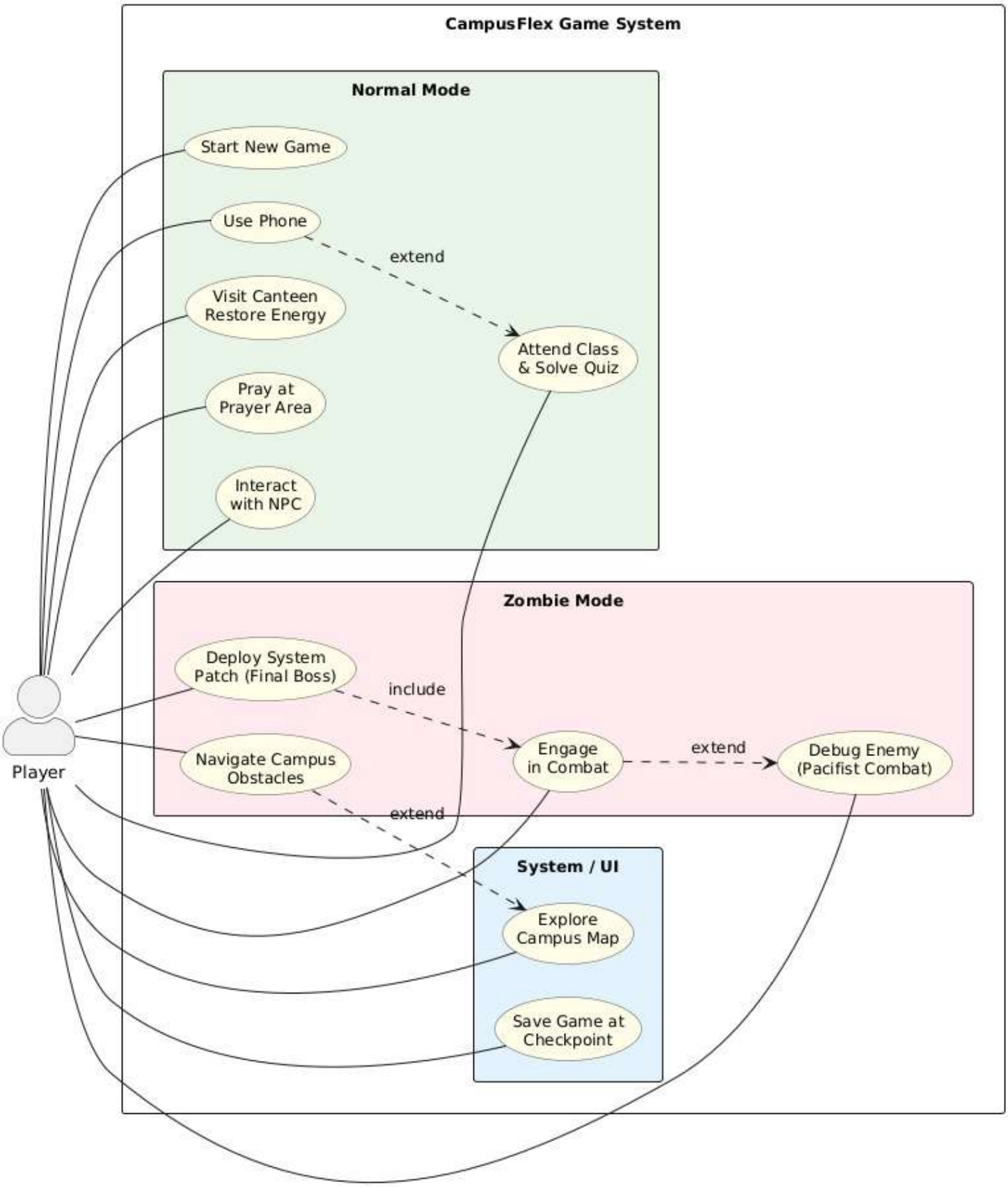

Use Case Summaries

A quick overview of all 12 use cases before the full tables.

#	Use Case	Summary
1	**Start New Game**	The player launches the game and kicks off a brand new session. The system shows an intro cutscene, sets all stats to their starting values (GPA, Energy, Stress, Karma), and drops the player at the C-Block entrance. This is the entry point for everything that follows.
2	**Attend Class & Solve Quiz**	The player walks into a classroom during scheduled hours and sits through a lecture. At one point, a quiz pops up. If the player answers correctly, their GPA goes up and they unlock knowledge that becomes useful in Zombie Mode. Cheating or skipping class has real stat consequences.
3	**Use Phone**	The player can pull out their phone at any time. Outside of class, scrolling social media kills a bit of time and Stress drops slightly but Karma takes a hit. During a quiz, using the phone counts as cheating and the attempt is thrown out with a bigger Karma penalty.
4	**Visit Canteen to Restore Energy**	When Energy runs low, the player heads down to the canteen on B1 and orders some food. The system bumps Energy back up and may ease Stress a little. If the player is already full, the system tells them so and nothing changes.
5	**Pray at Prayer Area**	The player visits the Prayer Area on the 2nd floor of C-Block and chooses to pray. This gives a Karma boost but can only be done once per in-game cycle to keep it from being exploited. If the player has already prayed, the prompt is disabled for that cycle.
6	**Interact with NPC**	The player walks up to classmates, professors, or staff and gets into a dialogue. The responses available depend on the current game mode and past interactions. Choosing a helpful or kind response increases Karma, while being dismissive brings it down - and NPCs will remember in Zombie Mode.
7	**Save Game at Checkpoint**	The player finds a glowing Save Point on the map and chooses to save. The system records everything i.e. current stats, location, flags, quiz results, and game mode to the database. If the save fails, no partial data is written and the player is told to try again.
8	**Explore Campus Map**	The player moves freely across the C and D Block campus zones. When approaching a zone boundary, the next area loads automatically. In Zombie Mode, certain paths are blocked and the system nudges the player toward the available alternate route.
9	**Engage in Combat**	When the player runs into a zombie, a combat screen kicks in. The player picks between two approaches: Debug (solve a challenge, keep Karma intact) or Terminate (deal damage directly, but Karma takes a hit). If the player's HP hits zero, they respawn at the last checkpoint.
10	**Debug Enemy (Pacifist Combat)**	Choosing Debug in a fight brings up a programming or logic challenge. Solve it correctly and the zombie is neutralized without any violence, earning a Karma boost. Get it wrong and the zombie bites back and the player can retry or fall back on Terminate.
11	**Navigate Campus Obstacles**	Some routes in Zombie Mode are sealed off. The player has to find alternate paths, pick up key items hidden around campus, and use

		them to unlock blocked doors. Guarded areas may require clearing a zombie first before the key can be grabbed.
12	**Deploy System Patch (Final Boss)**	The endgame. The player reaches the Research Lab, faces a corrupted AI in a final boss fight, and deploys a patch to stop the outbreak. The ending the player gets Pacifist, Evil, or Mixed is determined by the total Karma they've built up across both Normal and Zombie Mode.

Fully Dressed Use Cases

Use Case 1: Start New Game	
Use Case Name	Start New Game
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to start fresh with a new character and get into the game.

Main Success Scenario	
Actor Action	**System Response**
1. Player opens the game and lands on the Main Menu.	
2. Player selects 'New Game'.	
	3. System plays an intro cutscene that sets up the story.
	4. System sets all player stats to their starting values.
	5. System loads the FAST-NU campus map (C & D Block).
	6. Player spawns at the C-Block entrance.
	7. Normal Mode begins.

Extensions	- 2a. Player selects 'Load Game' instead - system loads from the last checkpoint. - 3a. Player skips the cutscene - system jumps straight to the spawn point.
Pre-conditions	The game is open and the Main Menu is on screen.
Post-conditions	A new game session is running. Stats (GPA, Energy, Stress, Karma) are at default values. Player is at C-Block Ground Floor. Normal Mode is active.

Use Case 2: Attend Class & Solve Quiz	
Use Case Name	Attend Class & Solve Quiz
Scope	CampusFlex Game System
Level	User-Goal

Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to sit through class, improve GPA, and pick up knowledge that carries into Zombie Mode.

Main Success Scenario	
Actor Action	**System Response**
1. Player walks up to a classroom and steps inside.	
	2. System plays the class attendance cutscene.
	3. Lecture content comes up as in-game dialogue.
	4. During one lecture, the system presents a quiz - either multiple choice or a short problem.
5. Player picks an answer or types a solution.	
	6. System checks the answer.
	7. If correct, GPA goes up and the lecture knowledge flag is set.
	8. Stats are updated and the player is returned to the campus map.

Extensions	- 4a. Player opens the phone during the quiz - Karma drops and the quiz auto-fails. - 5a. Player answers wrong - GPA still goes up a little, but no knowledge flag is applied. - 2a. Player skips class by not going in - GPA and Karma both take a quiet hit.
Pre-conditions	Normal Mode is running. The player is outside a classroom during scheduled class hours and hasn't attended this session yet.
Post-conditions	GPA has increased. If the quiz was answered correctly, a GPA bonus is applied and the lecture knowledge flag is set for Zombie Mode. If the player cheated, Karma has dropped.

Use Case 3: Use Phone	
Use Case Name	Use Phone
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Might want to kill some time or try to cheat on a quiz.

Main Success Scenario	
Actor Action	**System Response**
1. Player presses the phone button to open the Phone Menu.	
	2. System shows the Phone Menu (scroll social media, close phone, etc.)
3. Player picks 'Scroll Social Media'.	
	4. System plays a short animation of the player on their phone.
	5. Stats update: Stress -5, Karma -5.
	6. Phone closes and normal gameplay resumes.

Extensions	- 1a. Player opens the phone during an active quiz - system flags it as cheating, Karma drops by 15, and the current quiz attempt is cancelled. - 3a. Player picks 'Close Phone' - menu closes with no stat changes.
Pre-conditions	Normal Mode is active. The phone menu is accessible at any time via a key press.
Post-conditions	Outside a quiz: Stress drops a bit but Karma also drops. During a quiz: Karma takes a bigger hit and the quiz answer is wiped out.

Use Case 4: Visit Canteen to Restore Energy	
Use Case Name	Visit Canteen to Restore Energy
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Needs to top up Energy so they can keep moving around campus without slowing down.

Main Success Scenario	
Actor Action	**System Response**
1. Player heads to C-Block B1 and enters the Canteen.	
	2. System shows the Canteen menu prompt.
3. Player picks something to eat.	
	4. System plays a short eating animation.

	5. Player's Energy goes up.
	6. Player is sent back to the campus map.

Extensions	- 3a. Player leaves without ordering - no stat changes. - 2a. Player's Energy is already full - system responds: 'You're already full, bhai.' Nothing changes.
Pre-conditions	Normal Mode is running. Player is at the Canteen on C-Block B1. Energy is below max.
Post-conditions	Energy has been restored. Stress may have dropped slightly as a side effect.

Use Case 5: Pray at Prayer Area	
Use Case Name	Pray at Prayer Area
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to make a deliberate choice to build Karma.

Main Success Scenario	
Actor Action	**System Response**
1. Player goes to C-Block 2nd Floor and reaches the Prayer Area.	
	2. System shows an interaction prompt.
3. Player chooses to pray.	
	4. System plays a short prayer animation.
	5. Karma goes up.
	6. System marks the 'prayed this cycle' flag.
	7. Player is returned to the campus map.

Extensions	- 3a. Player decides not to pray - no stat change, player goes back to the map. - 2a. Player has already prayed this cycle - system says: 'You've already prayed today.' The prompt is grayed out.
Pre-conditions	Normal Mode is active. Player is at the Prayer Area on C-Block 2nd Floor and hasn't prayed yet this cycle.
Post-conditions	Karma has increased. The one-time flag for this cycle is set so the player can't repeat it for a free Karma boost.

Use Case 6: Interact with NPC	
Use Case Name	Interact with NPC
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to learn something, move the story along, or adjust Karma through a conversation. **NPC:** Reacts based on the current context and whether the player has spoken to them before.

Main Success Scenario	
Actor Action	**System Response**
1. Player walks near an NPC and a dialogue prompt appears.	
2. Player starts the conversation.	
	3. System pulls up the NPC's dialogue based on the current mode and past interactions.
4. Player picks a response.	
	5. System checks whether the choice affects Karma.
	6. Any stat changes are applied quietly in the background.
	7. Dialogue ends and the player is back on the map.

Extensions	- 4a. Player picks a rude or dismissive response - Karma drops. In Zombie Mode, this NPC might refuse to help or turn hostile. - 3a. NPC has a hidden Easter Egg line - system plays the sequence with no stat effect. - 2a. Player walks away without talking - nothing changes, NPC is still available.
Pre-conditions	Player is close enough to an NPC to trigger the prompt. The NPC hasn't been fully used up for this cycle.
Post-conditions	Dialogue has played out. Kind responses raise Karma; dismissive ones lower it. In Zombie Mode, past choices can change how NPCs behave toward the player.

Use Case 7: Save Game at Checkpoint	
Use Case Name	Save Game at Checkpoint
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to lock in their progress so nothing is lost if they quit or get knocked out.

Main Success Scenario	
Actor Action	**System Response**
1. Player walks up to a glowing Save Point on the map.	
	2. System asks: 'Save your progress?'
3. Player confirms.	
	4. System packages up the current game state: stats, location, flags, quiz results, and mode.
	5. System writes everything to the Oracle database.
	6. System confirms: 'Game Saved.'
	7. Player continues playing.

Extensions	- 3a. Player cancels - prompt closes, nothing is written. - 5a. The database write fails - system shows an error: 'Save failed. Try again.' No partial data is saved.
Pre-conditions	Player is standing on or right next to a Save Point.
Post-conditions	Current game state is stored in the database. Player can resume from this exact point when they load the game.

Use Case 8: Explore Campus Map	
Use Case Name	Explore Campus Map
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to roam the campus, reach objectives, and discover what's around them.

Main Success Scenario	
Actor Action	**System Response**
1. Player uses the directional controls to move across campus.	
	2. System keeps rendering the surrounding C and D Block zones.
3. Player reaches a zone boundary (e.g., the bridge from C-Block to D-Block).	
	4. System picks up the transition trigger.
	5. System loads the next zone.
6. Player keeps exploring in the new area.	

Extensions	- 3a. In Zombie Mode, a route is sealed (e.g., C-Block stairs are closed) - a barrier appears with a hint pointing toward the alternate path. - 5a. Zone fails to load - system shows an error and keeps the player where they are.
Pre-conditions	A game session is active (Normal or Zombie Mode). The player character is on the campus map.
Post-conditions	Player has moved to a new spot or loaded into a new zone. In Zombie Mode, blocked routes are enforced and the alternate path is signaled.

Use Case 9: Engage in Combat	
Use Case Name	Engage in Combat
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Needs to get past a zombie that's blocking the way to the upper-floor labs.

Main Success Scenario	
Actor Action	**System Response**
1. Player makes contact with a zombie.	
	2. System brings up the combat screen.
	3. System shows the enemy alongside the player's current HP.

	4. System presents two options: Debug (Pacifist) or Terminate (Brute Force).
5. Player makes their choice.	
	6. System follows the Debug path or applies Terminate damage.
	7. Enemy HP drops.
	8. If enemy HP hits zero, the fight ends, the path clears, and Karma is updated.
	9. Player is back on the campus map.

Extensions	- 7a. Player's HP hits zero - game over plays and the player respawns at the last checkpoint. - 4a. Player keeps picking Terminate - Karma keeps dropping. By the final boss, this shapes the ending.
Pre-conditions	Zombie Mode is active. Player has run into a zombie on the map. Player HP is above zero.
Post-conditions	Fight is over. If the enemy is defeated, the path is clear. If the player ran out of HP, they're back at the last checkpoint. Karma has changed based on how the fight went.

Use Case 10: Debug Enemy (Pacifist Combat)

Use Case Name	Debug Enemy (Pacifist Combat)
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to take down a zombie without hurting it - by solving a code or logic challenge instead.

Main Success Scenario

Actor Action	System Response
1. Player picks 'Debug' in a combat encounter.	
	2. System shows a programming or logic challenge (e.g., fix a broken snippet, answer a CS question).
3. Player types a solution or picks the right answer.	
	4. System checks the input.
	5. Correct - system plays a success animation and the zombie is pacified.

	6. Karma goes up.
	7. Fight ends, player is back on the map.

Extensions	- 4a. Player gets it wrong - the zombie bites back (minor HP loss). Player can retry or switch to Terminate. - 3a. Player takes too long - time penalty kicks in (HP loss) and the challenge resets, or the player is forced into Terminate.
Pre-conditions	A combat encounter is in progress. Player has chosen the 'Debug' option.
Post-conditions	If solved correctly, the zombie is neutralized without harm and Karma has increased. If wrong, the player took a hit and can still retry or switch tactics.

Use Case 11: Navigate Campus Obstacles

Use Case Name	Navigate Campus Obstacles
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Needs to find a way past blocked routes and locked doors to keep climbing toward the labs.

Main Success Scenario

Actor Action	System Response
1. Player tries to use a blocked route (e.g., C-Block stairs).	
	2. System shows a barrier with a message (e.g., 'Stairs are sealed. Find another way.')
3. Player takes the D-Block bridge as an alternate route.	
4. Player finds a locked door in D-Block that needs a key.	
5. Player heads to the key location (e.g., D-Block B2/B3 car park area).	
6. Player picks up the key.	
	7. System adds the key to the player's inventory and unlocks the door.
8. Player heads through the now-open route.	

Extensions	- 4a. Player tries to open the locked door without the key - system says: 'It's locked. You need to find the key.' - 6a. The key spot is guarded by a zombie - combat kicks in before the player can grab it.
Pre-conditions	Zombie Mode is active. Player has hit a blocked path or locked door on the way up.
Post-conditions	Player has the item or has found the alternate path. The blocked route is now open. The player can continue climbing toward the labs.

Use Case 12: Deploy System Patch (Final Boss)	
Use Case Name	Deploy System Patch (Final Boss)
Scope	CampusFlex Game System
Level	User-Goal
Primary Actor	Player
Stakeholders & Interests	**Player:** Wants to reach the Research Lab, beat the final boss, and end the outbreak.

Main Success Scenario	
Actor Action	**System Response**
1. Player enters the Research Lab at the top of D-Block.	
	2. System plays the final boss cutscene - a corrupted AI appears.
	3. Final boss fight starts (a bigger, climactic version of normal combat).
4. Player solves the final challenge or goes brute force.	
	5. Boss defeated - system shows the patch deployment interface.
6. Player confirms the patch.	
	7. System totals up Karma from all Normal and Zombie Mode choices.
	8. High Karma (Pacifist Ending): campus saved, player passes the semester, everyone celebrates.
	9. Low Karma (Evil Ending): campus saved, but player is expelled. Dark epilogue plays.
	10. Mixed Karma (Normal Ending): campus saved, bittersweet outcome.
	11. Credits roll, game returns to Main Menu.

Extensions	- 4a. Player fails the final challenge - HP drops. Player can retry or use Terminate at a Karma cost. - 7a. No checkpoint saved before the boss - system forces a retry from the lab entrance.
Pre-conditions	Zombie Mode is active. Player has reached the Research Lab on the top floor of D-Block. All obstacles have been cleared.
Post-conditions	Final boss is done. System has totalled Karma and picked an ending: Pacifist, Evil, or Normal/Mix. The game session is over.